

ROSACEA

Overview

Despite universal recognition, rosacea remains a controversial topic in dermatology, largely due to its uncertain pathophysiology and clinical variability. While the prototypical presentation—central facial erythema—is easily identifiable by both clinicians and the public, diagnostic confusion often arises when conditions such as photodamage, perioral dermatitis, postadolescent acne, and topical steroid-induced rosacea present with similar features.

Recent conceptual understanding has shifted away from a staged progression model toward a sub-type classification system that acknowledges variable severity and potential overlap among sub-types.

Rosacea is defined by persistent erythema of the central face—typically affecting the convex areas of the nose, cheeks, chin, and forehead—that lasts for months or longer.

Primary Features (may be present but not required for diagnosis):

- Flushing
- Papules
- Pustules
- Telangiectases

Secondary Features:

- Facial burning or stinging
- Edema
- Plaques

- Dry or rough skin appearance
- Phyma (e.g., rhinophyma)
- Peripheral flushing (involving scalp, ears, lateral face, neck, chest)
- Ocular manifestations (e.g., blepharitis, conjunctivitis, keratitis)

Note: Peripheral erythema may occur in rosacea but is also common in physiologic flushing and chronic sun damage; thus, careful clinical interpretation is necessary.

Sub-Type Classification

The National Rosacea Society (NRS) Expert Committee provisionally established four sub-types in 2002, based on predominant clinical features. These sub-types reflect common patterns rather than a fixed progression.

1. Erythematotelangiectatic Rosacea

- Persistent central facial erythema
- Frequent or persistent flushing
- Visible telangiectases
- May have skin sensitivity (burning, stinging)
- Often associated with dryness

Corresponds to Plewig-Kligman Stage I

2. Papulopustular Rosacea

- Central facial erythema with transient papules and pustules
- Resembles acne but lacks comedones
- May have oily skin
- Common in middle-aged women

Corresponds to Plewig-Kligman Stage II

3. *Phymatous Rosacea*

- Skin thickening and irregular surface nodularities
- Most commonly affects the nose (rhinophyma), but can involve chin, forehead, cheeks, eyelids
- Associated with long-standing disease
- More common in men

Corresponds to Plewig-Kligman Stage III

4. *Ocular Rosacea*

- Ocular symptoms precede or accompany skin findings in up to 50% of cases
- Symptoms: foreign body sensation, dryness, burning, stinging, photophobia, blurred vision
- Signs: lid margin telangiectasia, blepharitis, conjunctivitis, corneal involvement (keratitis)
- Can occur without obvious facial rosacea

The NRS classification emphasizes that these sub-types do not necessarily progress sequentially. Instead, they may coexist or appear independently in the same individual. A provisional grading system was also introduced to standardize assessment of lesion count, erythema severity, and phymatous changes.

Clinical evaluation should also consider the psychological, social, and occupational impact of rosacea, as well as individual response to therapy.

Epidemiology

- Rosacea affects all races but is most prevalent in fair-skinned individuals, particularly those of Northern or Eastern European descent.
- Peak onset: between 30 and 50 years of age.
- Slight female predominance for skin-predominant forms; phymatous changes are more common in males.

- Although underdiagnosed, rosacea is considered common in Caucasian populations.
- Cases are increasingly recognized in individuals of African and Asian descent, though presentation may differ due to variations in pigmentation and clinical recognition.

Triggers and Management Principles

Common Triggers:

- Temperature extremes (hot or cold)
- Sun exposure
- Wind
- Hot beverages
- Spicy foods
- Alcohol
- Emotional stress
- Exercise
- Cosmetics and topical irritants
- Menopausal flushing
- Medications causing vasodilation (e.g., calcium channel blockers, niacin)

General Management Strategies:

- **Sun protection:** Daily broad-spectrum sunscreen use (SPF 30+), physical barriers (hats, shade)
- **Trigger avoidance:** Patient education and identification of personal exacerbating factors
- **Gentle skin care:** Non-irritating cleansers, fragrance-free moisturizers, avoidance of abrasive products

Treatment Modalities:

- **Topical agents:** Metronidazole, ivermectin, azelaic acid, brimonidine (for erythema), oxymetazoline
- **Oral antibiotics:** Doxycycline (low-dose, anti-inflammatory), tetracyclines, macrolides (for inflammatory lesions)
- **Topical retinoids:** Off-label use; caution due to irritation potential
- **Isotretinoin:** Considered in severe or refractory cases, especially with papulopustular or phymatous features
- **Laser and light therapies:** Pulsed dye laser (PDL), intense pulsed light (IPL) for erythema and telangiectasia
- **Surgical or procedural interventions:** For phymatous changes (e.g., electrosurgery, dermabrasion, laser ablation)
- **Ocular management:** Artificial tears, lid hygiene, topical cyclosporine, oral doxycycline (low-dose)

Long-term control typically requires a combination of lifestyle modification, medical therapy, and periodic reassessment.